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ABSTRACT

The application of quantile regression offers a versatile and appealing approach for analyzing censored data, particularly within the accelerated failure time (AFT) model, where the focus lies on the significance of conditional quantile functions in regression analysis. The extension is achieved through the integration of well-established greedy algorithms—sure independence screening (SIS), tilting, and PC-simple—resulting in the development of Quant-SIS, Quant-Tilting, and Quant-PC techniques, respectively. These techniques prove to be adaptable, efficient, and consistent variable selection algorithms for high-dimensional datasets due to the inherent properties of sure independence, tilting correlation, and partial faithfulness. We compare the performance of the proposed methods with two competitive approaches from the existing literature. Through a comprehensive series of simulation studies encompassing diverse scenarios including varying collinearity levels among covariates, levels of censoring, and quantiles—we demonstrate their efficacy. Additionally, we apply the proposed methods to real-world microarray data from Diffuse Large B-cell Lymphoma (DLBCL) patients. This application showcases the ability of our techniques to accurately identify genes associated with the survival time of DLBCL patients. The results indicate a substantial enhancement in performance, as the modified quantile regression techniques for censored data significantly outperform existing methods across a wide spectrum of cases.

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Accelerated failure time models; Greedy algorithms; High dimensional censored data; Variable selection

1. Introduction

Koenker and Bassett (1978) developed quantile regression for pursuing a compatible analysis of classical linear regression. Quantile regression focuses at evaluating the conditional median or other quantiles of the response variable on the other hand least square procedures only estimate the conditional mean of the response variable given a specific value of the predictors. Nowadays quantile regression slightly polished with different algorithm has also become one of the tool in high dimensional data analysis. Because in this era of data science the whole world invest enormously for gathering and operating data form all sources. Medicine, pharmaceuticals, banking, e-commerce, finance, insurance, retailing, manufacture, telecommunication, transportation, gaming etc are few familiar sources of databases in the world. Due to updating data every day the dimension of data sets get bigger and bigger. High dimensionality basically refers that the number of variables in the data set will be larger than the number of observations. Variable selection is one of the fundamental thing to do in high dimensional data analysis because one have to identify the relevant variables from a hyperspace which is a mixture of the relevant and irrelevant variables.

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Researchers face many challenges dealing these ultra high dimensional data sets. Because when the dimension of a data set increases number of predictors on that data set also increase often more quickly which indicates that there is a chance of immense sparsity associated with such high dimensional predictors Khan (2018). The correlation between variables happened be so high that cause various problems in modeling. Due to high multicollinearity between variables the irrelevant variables tend to have significant effect on response variable. Different techniques and algorithms have been used by the researchers for variable selection in high dimensional data set. Tibshirani (1996) developed a regression shrinkage and variable selection method named "Lasso" that minimizes the residual sum squares. Agrawal *et al.* (1998) established clustering based algorithm to find clusters embedded in subspaces of high dimensional data that identifies dense clusters in subspaces of maximum dimensionality. Yu and Yang (2001) developed a direct and exact linear discriminant analysis (LDA) for diminishing the dimensionality of high dimensional data. McLachlan, Peel, and Bean (2003) focused on mixtures of factor analyzers from the perspective of a method for model-based density estimation from high-dimensional data. Yu and Liu (2003) introduced a method which can identify relevant features as well as redundancy among relevant features without pairwise correlation analysis. Zou and Hastie (2005) developed elastic net, a new regularization and variable selection method which encourages a grouping effect when the number of predictors is much bigger than the sample size.

Many researchers implemented quantile regression with different penalization technique or machine learning algorithm in high dimensional data analysis. Meinshausen (2006) developed a quantile regression methods with one of the popular machine learning algorithm random forest. Belloni and Chernozhukov (2011) introdcued l_1 penalized quantile regression which penalized the l_1 norm of regression coefficients. He, Wang, and Hong (2013) developed a quantile adaptive model free algorithm for screening variables from high dimensional heterogeneous data. Zheng, Gallagher, and Kulasekera (2013) established a penalization structure for quantile regression in high dimensional framework. Tung *et al.* (2014) introduced bias corrected quantile regression forests for high dimensional data. Park, He, and Zhou (2017) developed a dantzig type penalization for multiple quantile regression with high dimensional covariates.

In survival analysis, for ultra high dimensional censored data, one of the major problems is incompleteness due to censoring. So if we have ultra high dimensional censored data and we want to identify the dynamic predictors, the first thing to do is to deal with the censoring. Tibshirani (1997) developed a new method for variable selection and shrinkage in Cox's proportional hazards model. Faraggi and Simon (1998) introduced a Bayesian variable selection procedure for censored data. Fan and Li (2002) extended the non-concave penalized likelihood approach further to the Cox proportional hazards model and the frailty model. Gui and Li (2005) used the L_1 penalized estimation of the Cox model. Johnson, Lin, and Zeng (2008) developed a general strategy for variable selection in semi parametric regression models by penalizing appropriate estimating functions. Recently Xu, Leng, and Ying (2010) developed a rank-based variable selection procedure for the semi parametric accelerated failure time model with censored observations. Khan (2013) developed two variable selection procedures based on regularized weighted least squares and another is adaptive elastic net for accelerated failure time models. Khan and Shaw (2016) developed a variable selection approach which is a mixture of l_1 and l_2 norm penalties AFT models. Xia *et al.* (2016) also employed Stute's estimator for nonparametric variable screening and selection. Khan and Shaw (2019) established another variable selection procedure for AFT models based on the Buckley–James method and the Dantzig selector, notably used in the filed of variable selection and survival analysis. Variable selection with stability selection for regularized models under AFT models has been discussed in Khan, Bhadra, and Howlader (2019). Ranking based variable selection method for AFT models has been discussed in Khan and Akhter (2024). Recently, Salerno and Li (2023) discussed all high dimensional methods for censored data including their applications.

Quantile regression also have heen embedded in survival analysis with ultra high dimensional censored data. One of the major problem in dealing ultra high dimensional censored data is in completeness due to censoring. Portnoy (2003) proposed a censored regression quantile method. Lu and Zhang developed a variable selection approach through censored median regression using the inverse probability weighting scheme of Bang and Tsiatis (2002). This method requires the restrictive unconditional independence assumption between survival and censoring times. Wang, Zhou, and Li (2013) developed a new adaptive lasso based variable selection procedure for quantile regression with censored outcomes. Reich and Smith (2013) developed a semi parametric quantile regression model for censored survival data.

Censored quantile regression (CQR) techniques addressing randomly censored survival data have been extensively examined in the context of finite p scenarios (Peng and Huang (2008), Wang and Wang (2009), Zheng, Peng, and He (2015), Zheng, Peng, and He (2018) etc.). Leng and Tong (2013) established a censored quantile regression estimator motivated by an unbiased estimating equations. But this method requires the independence assumption of the survival time and the censoring time given the covariates.

The algorithms that have been used to deal ultra high-dimensional censored data, none of them have implemented greedy algorithms in their model framework. The greedy algorithms can deal with the correlation among predictors and reduce the dimension at a lower space very fast compared to other selection methods. If quantile regression is embedded with greedy algorithms and penalization an outstanding approach for variable selection can be developed. The main goal of this study is to modify the quantile regression through greedy algorithms for high-dimensional censored data so that the relevant predictors will be selected accurately. Fan and Lv (2008) established the sure independent screening procedure (*SIS*) to diminish the dimensionality of ultra high dimensional data. Cho and Fryzlewicz (2012) proposed a procedure called *tilting* which yields an alternative between the use of marginal and tilted correlation for each variable. Bühlmann, Kalisch, and Maathuis (2010) proposed the new concept of partial faithfulness and use it to infer associations between the covariates and response. Using partial faithfulness, they developed a simplified version of the *PC-simple algorithm*. In this study we illustrate how the quantile regression can be embedded with SIS, tilting and *PC-simple algorithm*.

2. Methods

2.1. Censored quantile regression and variable selection

Generally two models namely the accelerated failure time model and Cox proportional hazard (Cox PH) model are popularly used models in survival analysis. This study uses the AFT model over the Cox PH model since the AFT model has an intuitive physical interpretation and would be a useful alternative to the Cox model in survival analysis [Wei (1992)].

Let, T_i and C_i be the logarithm of survival and censoring time respectively for $i = 1, 2, ..., n$. So our observed logarithm follow up time $Y_i = \min(T_i, C_i)$ and $\delta_i = I(T_i \leq C_i)$ be the censoring indicator. Also let X_i be the covariate vector. The AFT model is defined as

$$Y_i = \beta_0 + x_i^T \beta + \epsilon_i, \quad i = 1, 2, ..., n, \tag{1}$$

where β_0 is the intercept, $\beta \in \mathbb{R}$ is the regression coefficients and ϵ_i is the error term. If survival time T_i follows the Weibull, the log normal and the log logistic distribution then ϵ_i follows the extreme value, the normal and the logistic distribution respectively.

The quantile regression model for censored data model as expressed in Equation (1) can be expressed as

$$Y_i = x_i^T \beta(\tau) + \epsilon_i(\tau), \tag{2}$$

where $0 < \tau < 1$ is given quantile level of interest, $\beta(\tau)$ is unknown quantile coefficient vector and $\epsilon_i(\tau)$ is τth quantile of the random error term but doest not depend on x_i. Without loss of generality, we assume that the first element of x_i is 1, corresponding to the intercept. Our main objective is to select important variables that have non-zero effect on the τ-th conditional quantile of T in the model.

For variable selection, we use the idea of redistribution-of-mass in censored quantile regression as proposed in Wang, Zhou, and Li (2013). Efron (1967) proposed a simple algorithm for deriving the Kaplan-Meier estimator by redistributing the mass of each censored observation uniformly to observations on the right when censored data is present without covariates, that is, $p = 1$ in Equation (2). That is, under the regression setup, this means to redistribute the probability masses $P(T_i > C_i|C_i, x_i)$ of censored cases to observations on the right. Let $F(t|x) = P(T < t|x)$ denote the conditional distribution function of T given x, and take $\pi_i = F(C_i|x_i)$ as the conditional probability for the i-th subject not to be censored. Consider an ideal scenario where the π_i are known. Then $\beta(\tau)$ can be estimated by minimizing a weighted quantile objective function with respect to β.

$$L(\beta(\tau), w) = \sum_{i=1}^{n} w_i \rho_\tau(Y_i - x_i^T \beta(\tau)) + (1 - w_i)\rho_\tau(Y^{+\infty} - x_i^T \beta(\tau)), \tag{3}$$

where $\rho_\tau(u) = u(\tau - I(u < 0))$ is the check function, $Y^{+\infty}$ is any value sufficiently large to exceed $\beta(\tau)$ for all i, and

$$w_i = \begin{cases} 1, & \text{if } \delta_i = 1 \text{ (uncensored)}, \\ 0, & \text{if } \delta_i = 0 \text{ and } \pi_i > \tau, \\ \dfrac{\tau - \pi_i}{1 - \pi_i}, & \text{if } \delta_i = 0 \text{ and } \pi_i \le \tau. \end{cases}$$

In practice, the masses for redistribution, $1 - \pi_i = 1 - F(C_i|x_i)$, are unknown, and a variety of attempts have been made to estimate them. We followed the index-based procedure to obtain a nonparametric estimation of π_i for multivariate covariates as proposed in Wang, Zhou, and Li (2013). Hence, the underlying framework can be treated as the adaptive censored quantile regression framework through the AFT model. It is conjectured that the weighted quantile regression estimator $\hat{\beta}(\tau)$ is consistent and asymptotically normal. The sketch of proof is given in Appendix. This proposed method has been compared with a latest CQR method (Zheng, Peng, and He 2018) later in the analysis section.

CQR does not assume a specific parametric form for the distribution of survival times, unlike the AFT model, which assumes a particular distribution (e.g. Weibull, log-normal) that may not be suitable for all datasets. CQR is robust to outliers and can handle heteroscedasticity (non-constant variance) more effectively than parametric models like AFT, making it ideal for datasets with extreme values or varying dispersion. Additionally, CQR allows for the analysis of different points (quantiles) of the conditional distribution of the response variable, providing a more comprehensive view of the relationship between covariates and the outcome across the entire distribution, rather than focusing solely on the mean or median.

2.2. Quantile regression with greedy algorithm

A greedy algorithm makes greedy choices at each step to ensure that the objective function is optimized perfectly. Greedy algorithm is naturally quite easier to come up with the optimum solution than others. Of interest, we would like to optimize the above objective function Equation (3) with a set of greedy algorithms in literature. We consider SIS (Fan and Lv 2008), tilting (Cho and Fryzlewicz 2012) and PC-simple algorithm (Bühlmann, Kalisch, and Maathuis 2010) to be

embeded with the above objective function leading "Quant-SIS", "Quant-Tilting" and "Quant-PC" algorithms respectively. It can be noted that all other tuning parameters of the methods were tuned through a cross-validation method.

The introduction of greedy methods is crucial when faced with the task of optimizing an objective function, whether it requires maximization or minimization at a specific point. Formulating a greedy algorithm, or even multiple alternatives, for a given problem is notably straightforward. Greedy strategies are intuitive and often involve making locally optimal choices. This methodical approach allows researchers to efficiently select variables, and though proving the correctness of greedy algorithms can be intricate, their suitability for solving optimization challenges, like variable selection, is indisputable.

2.2.1. Quant-SIS

The sure independent screening (Fan and Lv 2008) is a screening procedure that can select all significant variables with probability tending to one. The SIS reduces the full model containing p variables into a sparse model, $M_* = \{1 \le i \le p : \beta(\tau)_i \ne 0\}$ with s variables. The remaining $p - s$ variables may have effects on the response through their correlation with the s important variables. Let $v = (v_1, v_2, ..., v_p)^T$ be a p-vector obtained by the component wise regression, that is

$$v = X^T y. \tag{4}$$

For any given $\gamma \in (0, 1)$, Fan and Lv (2008) assigned the p vector v in a decreasing order. The sub model is then written as

$$M_\gamma = \{1 \le i \le p : \ |v_i| \text{ is among the first } [\gamma n] \text{ largest of all}\},$$

where $[\gamma n]$ denotes the integer part of γn. That is the way of reducing the full model $\{1, 2, ..., p\}$ down to a sub model M_γ with size $d = [\gamma n] < n$. Thus the original goal of estimating the sparse p-vector $\beta(\tau)$ in Equation (3) reduces to estimating a sparse d-vector $\beta(\tau) = \{\beta(\tau)_1, ..., \beta(\tau)_d\}$ on the basis of the sub model M.

This model is termed as "Quant-SIS" under which the quantile regression is implemented with specific τth quantile level. According to Fan and Lv (2008) SIS is followed by three penalty functions that exhibit three properties– sparsity, unbiasedness and continuity. One of the penalties is smoothly clipped absolute deviation (SCAD), as proposed by Fan (1997), is a continuously differentiable penalty function called the SCAD penalty and which is defined by

$$p_\lambda(|\beta(\tau)|) = \lambda\left\{I(|\beta(\tau)| \le \lambda) + \frac{(a\lambda - |\beta(\tau)|)_+}{(a-1)\lambda} I(|\beta(\tau)| > \lambda)\right\} \quad \text{for some } a > 2 \tag{5}$$

that corresponds to a quadratic spline function with knots at λ and $a\lambda$. The SIS followed by this SCAD penalty is named as SIS-SCAD. Combining SIS-SCAD and quantile regression we have referred the model as "Quant-SIS-SCAD".

Second penalty function is the minimax concave penalty (MCP) which was introduced by Zhang (2007). According to Fan and Lv (2008), MCP transforms the flat part of the derivative of the SCAD to the origin and is given by

$$p'_\lambda(|\beta(\tau)|) = \frac{(a\lambda - |\beta(\tau)|)_+}{a}$$

which minimizes the maximum of the concavity. SIS followed by this MCP is named as SIS-MCP. Combining quantile regression and SIS-MCP we named the model as "Quant-SIS-MCP".

The last penalty function is adaptive Lasso which was proposed by Zou (2006). The idea of such penalty is to use an unweighted l_1 penalty. Specially Zou (2006) introduced the following penalization term

$$\lambda \sum_{j=1}^{d} u_j \left| \beta(\tau)_j \right|, \tag{6}$$

where $\lambda \geq 0$ is the Lasso regularization parameter and $u = (u_1, ..., u_d)^T$ is a known weight vector. SIS followed by this penalization is named as SIS-Lasso. Combining this penalty and quantile regression we referred the model as "Quant-SIS-Lasso".

The SCAD penalty maintains the penalization rate of Lasso for smaller coefficients, but it gradually reduces the rate of penalization as the absolute value of the coefficient grows larger. Asymptotically, the SCAD penalty provides $\sqrt{n}$-consistent estimates (assuming an appropriate rate for η) and exhibits oracle properties (if $\sqrt{n}\lambda \to 0$ and $n\lambda \to \infty$) Salerno and Li (2023). Hence, we conjecture that the proposed techniques–"Quant-SIS-SCAD" and"Quant-SIS-Lasso" are expected to provide consistent estimates which exhibit oracle properties.

2.2.2. Quant-tilting

The tilting algorithm (Cho and Fryzlewicz 2012)] can successfully discriminate between relevant and irrelevant variables thus can be applied as a tool for variable selection. Under tilting, each column of X is assumed to have a unit norm and thus the sample correlation matrix of X is defined as $C = X^T X = (c_{j,k})_{j,k=1}^{p}$. If $\mathbb{D}$ be a subset of the index set $\mathbb{J} = \{1, 2, ..., j\}$, then $X_{\mathbb{D}}$ denotes the $n \times |\mathbb{D}|$ sub matrix of X with X_j, $j \in \mathbb{D}$ denotes its column. Similarly $\beta(\tau)_{\mathbb{D}}$ denotes the sub vector of $\beta(\tau)$. The process of tilting is to define the tilted correlation between each variable and the response. Then under the quantitle regression framework the marginal correlation between each variable X_j and y, as proposed in Cho and Fryzlewicz (2012), has the following structure

$$X_j^T y = X_j^T \left(\sum_{k=1}^{p} \beta(\tau)_k X_k + \epsilon \right) = \beta(\tau)_j + \sum_{k \in \mathbb{S} \setminus \{j\}} \beta(\tau)_k X_j^T X_k + X_j^T \epsilon. \tag{7}$$

Cho and Fryzlewicz (2012) examined the middle part of the Equation (7) and viewed it as a "bias" term. The main idea of tilting is that, transforming each X_j in such a way that the middle part for the transformed X_j is zero or negligible, not diminishing the effect of j-th covariate to the response. Cho and Fryzlewicz described several properties of such tilted correlation. Quantile followed by tilting is named as "Quant-Tilting".

2.2.3. Quant-PC

The simplified version of PC-simple algorithm is developed by Bühlmann, Kalisch, and Maathuis (2010). $X = (X^{(1)}, ..., X^{(p)}) \in \mathbb{R}^{|}$ be the vector of covariates with $E(X) = \mu_X$ and $cov(X) = \Sigma_X$. Also let $\epsilon \in \mathbb{R}$ with $E(\epsilon) = 0$ and $var(\epsilon) = \sigma^2 > 0$, and ϵ is uncorrelated with $X^{(1)}, ..., X^{(p)}$. And they defined $Y \in \mathbb{R}$ by the following random design linear model,

$$Y = \zeta + \sum_{j=1}^{p} \beta_j X^{(j)} + \epsilon \tag{8}$$

for some parameters $\zeta \in \mathbb{R}$ and β be the vector of coefficients with elements β_j. We assume that $E(Y^2) < \infty$ and $E\{(X^{(j)})^2\} < \infty$ for $j = 1, ..., p$.

The goal would be to identify the significant set $A = \{j = 1, 2, ..., p; \beta_j \neq 0\}$ based on the sample independent observation $(X_1, Y_1), ..., (X_n, Y_n)$ that are distributed as X,Y. We denote the effective dimension of the model, that is, the number of nonzero β_j's by peff $= |A|$. PC-simple algorithm identifies the significant variables using partial faithfulness. The distribution of (X, Y) is said to be a

partially faithful if for every $j \in \{1, ..., p\}$, $\rho(Y, X^{(j)}|X^{(S)}) = 0$ for some $S \subseteq \{j\}^C$ then $\rho(Y, X^{(j)}|X^{(j)^C}) = 0$. That is, partial faithfulness is that the correlation between the relevant predictor and the response variables is uncorrelated with the irrelevant variables. So embedding PC-simple algorithm with quantile regression model as define in Equation (3) leads to method "Quant-PC".

3. Numerical analysis

We compare the performance of proposed methods with quantile regression forest (QRF) as developed by Meinshausen (2006) and a high-dimensional censored quantile regression (CQR) method as developed by Zheng et al. The QRF gives a non-parametric and accurate way of estimating conditional quantiles for high-dimensional predictor variables and shows to be consistent. The author shows through numerical examples that the algorithm is competitive in terms of predictive power. The algorithm is shown to be consistent. Numerical examples suggest that the algorithm is competitive in terms of predictive power. The algorithm of QRF for computing the estimate $\hat{F}(\tilde{y}|\tilde{x})$ can be summarized as:

i. Grow k trees $Tr(\vartheta)$, $b = 1, ..., \kappa$, as in random forests. However, for every leaf of every tree, take note of all observations in this leaf, not just their average.
ii. For a given $X = x$, drop x down all trees. Compute the weight $u_i(\tilde{x}, \vartheta_b)$ of observation $i \in 1, ..., n$ for every tree. Compute weight $u_i(\tilde{x})$ for every observation $i \in 1, ..., n$ as an average over $u_i(\tilde{x}, \vartheta_b)$, $b = 1, ..., \kappa$.
iii. Compute the estimate of the distribution function $\hat{F}(\tilde{y}|\tilde{x})$.

The high-dimensional CQR method is based on the conditional quantiles over a continuum of quantile indices. This CQR method initially introduces an L_1 penalty, akin to Lasso, into the estimating equation for CQR through a stochastic integral approach, yielding a uniformly consistent estimator. Subsequently, it addresses bias introduced by the L_1 penalty in the second step, employing an adaptive Lasso type penalty. This refined approach enhances both estimation efficiency and model selection consistency in the resulting estimator. This CQR method leverages (i) valuable information across quantiles to enhance the resilience of variable selection, (ii) overall sparsity more succinctly, and (iii) encompassing localized quantile regression considerations.

3.1. Simulation studies

The covariates are generated using Uniform distribution $U(0, 1)$. The correlated covariates are generated using Cholesky factorization method. The Cholesky decomposition of a Hermitian positive-definite matrix A is a decomposition of the form

$$A = LL^*,$$

where L is a lower triangular matrix with real and positive diagonal entries. Matrix of the covariates generated using Uniform distribution, is multiplied by the positive definite matrix A. This will result into a matrix of correlated covariates. We consider correlation level 0.5 between covariates. The logarithm of the survival time is generated from

$$Y_i = \beta_0 + X_i^{'}\beta + \sigma\epsilon_i, \tag{9}$$

where σ is the noise and the error ϵ_i follows a particular distribution function $f(.)$ (e.g. extreme value or normal or logistic distribution). Both censored observation and survival time have been generated from log-normal distribution and two censoring rates $P_{\%} = 30\%$ and 50% have been considered.

Performance of the proposed variable selection methods are evaluated by false negatives (FN), false positives (FP) and FN+FP. Generally, FN is defined as the number of relevant variables incorrectly identified as non-relevant and FP is defined as the number of non-relevant variables incorrectly identified as relevant all averaged over 100 simulation runs under both censoring rates. The FN+FP indicates the total error occurred by the methods. Predictive performance of the method for real data analysis is done by the mean squared error.

3.1.1. Example I: when $p = n$

A moderate dimension with equal number of variables and sample size i.e. $n = 200$ and $p = 200$ has been considered in this example. Five coefficients of the variables $X_1, ..., X_5$ are considered non zero (say, p_α) with their corresponding coefficients are set to 10 and the remaining β's are considered zero.

Both correlated and uncorrelated covariates are considered. As the quantile regression has been embedded with greedy algorithms, so a specific τ-th quantile needs to be mentioned. We use two sets of τ: 0.25 and 0.6. The logarithm of survival time is generated from Equation (9), with ϵ_i follows $N(0, 1)$ distribution and fixing σ to be 1. All methods including quantile regression forests and high-dimensional quantile censored regression models have been implemented to the data set and the results are summarized in Table 1.

Table 1 reports the results for $\tau = 0.6$ and Tables A1 and A2 in Appendix report the results for $\tau = 0.25$ and $\tau = 0.50$. Table 1 demonstrates that under 30% censoring for both uncorrelated and correlated data, the proposed methods exhibit an FN+FP rate of 0, except for Lasso (0.003), while QRF (0.060) and CQR (0.011) show slightly inferior performance. Similar results were observed for correlated data. Under 50% censoring, Quant-tilting and Quant-PC outperform the others. The remaining methods perform comparably for both data types, with slightly diminished performance in the case of correlated data. The overall performance of the proposed methods aligns well with the QRF and high-dimensional CQR methods. Thus, the embedded methods demonstrate substantial efficacy, on par with existing methods in the literature. We observed highly similar trends in the results when the quantile level was set to 0.25 and 0.50 under the Example 1 configuration. Furthermore, the results for the quantile level of 0.50 closely matched those obtained when the quantile level was set to 0.60. We noticed similar results under low SNR (see Table D1 in appendix).

3.1.2. Example II: when $p \gg n$

In this instance, the number of variables is amplified to be twice the sample size, i.e. $n = 400$, and five times the covariate size, i.e. $p = 1000$, compared to the initial example. The coefficients

Table 1. Summary results of Example-I: $n = 200$ and $p = 200$, $p_\alpha = 5$ and $\tau = 0.60$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.001	0.000	0.000	0.000	0.060	0.010
	0	FP	0.000	0.002	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.003	0.000	0.000	0.000	0.060	0.011
		FN	0.000	0.001	0.000	0.000	0.000	0.000	0.010
	0.5	FP	0.000	0.001	0.000	0.000	0.000	0.009	0.002
		FN+FP	0.000	0.002	0.000	0.000	0.000	0.009	0.012
50%		FN	0.000	0.002	0.000	0.050	0.000	0.000	0.011
	0	FP	0.000	0.003	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.005	0.000	0.050	0.000	0.000	0.012
		FN	0.030	0.010	0.020	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.002	0.000	0.000	0.000	0.010	0.011
		FN+FP	0.030	0.012	0.020	0.000	0.000	0.010	0.011

Table 2. Summery results of example II: $n = 400$ and $p = 1000$, $p_\alpha = 10$ and $\tau = 0.60$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0	FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN	0.000	0.000	0.001	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.000	0.001	0.000	0.000	0.000	0.000
50%		FN	0.000	0.001	0.000	0.000	0.000	0.008	0.000
	0	FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.002	0.000	0.000	0.000	0.008	0.000
		FN	0.000	0.001	0.000	0.000	0.000	0.002	0.000
	0.5	FP	0.000	0.010	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.011	0.000	0.000	0.000	0.002	0.000

of the first 10 variables ($p_\alpha = 10$) are set to 10. Other correlation and censoring settings remain analogous to Example I, and the outcomes are summarized in Table 2 here for $\tau = 0.6$ and in Table B1 for $\tau = 0.25$ including Table C1 for $\tau = 0.50$ in Appendix. Across both correlated and uncorrelated datasets, and for both levels of censoring, the FN, FP, and FN+FP rates are all observed to be 0 for all methods, except for the proposed Lasso and its counterpart QRF methods. These proposed methods exhibit superior performance compared to the Lasso and QRF techniques in high-dimensional scenarios. Therefore, as the covariate size p increases reasonably, the proposed methods exhibit performance similar to existing methods in the literature, regardless of variations in censoring and correlation within the datasets. We observed highly similar trends in the results when the quantile level was set to 0.25 and 0.50 under the Example 2 configuration. Furthermore, the results for the quantile level of 0.50 closely matched those obtained when the quantile level was set to 0.60.

3.1.3. Example III: when $n \ll p$ with blocking effect

We have considered ($n = 100$ and $p = 200$) with blocking effect which is common to many high-dimensional data including microarray. Covariates are generated from $MVN(0, \Sigma)$ and divided into four blocks. Each of the first 3 blocks contain 5 covariates and the 4th block contains 185 covariates. We have assumed that only 1st and 3rd block contain relevant variables whose coefficients β's are set as 10 while for the rest as 0.

Note that each block has different correlation structure and the variance-covariance matrix of the covariates is set as

$$\Sigma = \begin{bmatrix} \Sigma_0 & & & \\ & \Sigma_0 & 0.2J & \\ & 0.2J & \Sigma_0 & \\ & & & \Sigma_0 \end{bmatrix},$$

where Σ_0 is a 15×15 matrix with diagonal and off diagonal elements 1, and 0.7 respectively; J is a 15×15 matrix with elements 1, within and between block correlation is set as 0.7 and 0.2 respectively. Performance measure (PM) statistic is used to quantify the performances of the methods for each block. The PM calculates proportion of the total number of selected relevant variables to the total number of true relevant variables considering 100 simulation runs. The results are summarized in Table 3.

When considering the quantile level $\tau = 0.25$, the proposed methods precisely identify the pertinent variables and disregard the irrelevant ones under both 30% and 50% censoring levels (PM1 = PM3 = 1 and PM2 = PM4 = 0). Only minimal distortion is observed in the performance

Table 3. Performance summary of example III: $n = 100$ $p = 200$ and $p_\alpha = 10$ with blocking effect and $\tau = 0.25, 0.50, 0.60$.

	$\tau = 0.25$											
	$P_{\%} = 30$						$P_{\%} = 50$					
Method	PM-1	PM-2	PM-3	PM-4	FN	FP	PM-1	PM-2	PM-3	PM-4	FN	FP
Quant-SIS-Scad	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-SIS-Lasso	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-SIS-MCP	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-tilting	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.99	0.00	0.00	0.00
Quant-PC	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
QRF	1.00	0.02	1.00	0.02	0.00	0.02	1.00	0.01	1.00	0.02	0.00	0.01
Hd-CQR	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.01	0.00	0.01
	$\tau = 0.50$											
	$P_{\%} = 30$						$P_{\%} = 50$					
Method	PM-1	PM-2	PM-3	PM-4	FN	FP	PM-1	PM-2	PM-3	PM-4	FN	FP
Quant-SIS-Scad	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-SIS-Lasso	0.99	0.01	1.00	0.01	0.00	0.00	1.00	0.00	0.99	0.00	0.00	0.00
Quant-SIS-MCP	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-tilting	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.99	0.00	0.00	0.00
Quant-PC	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
QRF	1.00	0.01	1.00	0.01	0.00	0.02	1.00	0.01	1.00	0.01	0.00	0.01
Hd-CQR	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.01	0.99	0.00	0.00	0.00
	$\tau = 0.60$											
	$P_{\%} = 30$						$P_{\%} = 50$					
Method	PM-1	PM-2	PM-3	PM-4	FN	FP	PM-1	PM-2	PM-3	PM-4	FN	FP
Quant-SIS-Scad	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-SIS-Lasso	0.99	0.01	1.00	0.01	0.00	0.00	1.00	0.00	0.99	0.00	0.00	0.00
Quant-SIS-MCP	0.99	0.00	0.99	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Quant-tilting	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.99	0.00	0.00	0.00
Quant-PC	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
QRF	1.00	0.02	1.00	0.02	0.00	0.02	1.00	0.01	1.00	0.02	0.00	0.01
Hd-CQR	1.00	0.00	0.99	0.00	0.00	0.00	1.00	0.01	0.99	0.00	0.00	0.01

of the Quant-tilting and QRF methods under 50% censoring. Specifically, FN and FP rates are both 0 when considering all blocks together. For QRF, PM1 $=$ PM3 $= 1$, while PM2 $=$ PM4 $=$ 0.02 under 30% censoring, and these values become 0.01 and 0.02 under 50% censoring. Additionally, we find $FN = 0$ but $FP = 0.01$ and 0.02 under 30% and 50% censoring rates, respectively. Thus, the QRF can accurately identify relevant variables but is less effective in disregarding irrelevant ones compared to the proposed methods. However, at $\tau = 0.50, 0.6$, the proposed methods, along with the QRF and Hd-CQR methods, exhibit a slight decline in performance.

3.2. Analysis of DLBCL data of Rosenwald et al. (2002)

The performance of proposed methods and QRF have been evaluated by a real dataset known as Diffuse large-B-cell lymphoma (DLBCL) of Rosenwald *et al.* (2002). The DLBCL consists of gene expression values of 7399 genes from 240 patients. The survival time was recorded from each patient, ranging between 0 and 21.8 years. Median survival time for these patients is 2.8 years. At the end of the follow up time 133 patients were uncensored and 102 patients were alive or censored. The comprehensive information and the data will be available in https://llmpp.nih.gov/DLBCL/. This data was divided by it's author into two dataset, training and test data consisting 160 and 80 patients respectively. The 7399 gene expression values and the survival time of lymphoma patients have been considered as predictors and response variable respectively. The proposed methods including QRF and Hd-CQR have been implemented to the training data set for identifying relevant genes under one quantile level $\tau = 0.6$. Result are summarized in Table 4.

Table 4. Performance of the methods for DLBCL example.

Methods	Quant-SIS-SCAD	Quant-SIS-Lasso	Quant-SIS-MCP	Quant-Tilting	Quant-PC	QRF	Hd-CQR
Quant-SIS-SCAD	9	7	7	5	7	7	6
Quant-SIS-Lasso	7	11	11	5	10	9	8
Quant-SIS-MCP	7	11	11	5	10	9	8
Quant-Tilting	5	5	5	12	8	9	8
Quant-PC	7	10	10	8	14	12	10
QRF	7	9	9	9	12	18	9
Hd-CQR	6	8	8	8	10	9	15

Table 5. Mean squared error, median of fitted values and *p*-value of log rank test by the methods.

Methods	MSE	Median of Fitted Values	*p*-value
Quant-SIS-SCAD	3.95	0.139	< 0.001
Quant-SIS-Lasso	4.20	−0.279	< 0.001
Quant-SIS-MCP	3.66	−0.009	< 0.000
Quant-Tilting	3.82	0.000	< 0.000
Quant-PC	4.70	−0.210	< 0.000
QRF	4.78	0.000	< 0.000
Hd-CQR	4.00	−0.051	< 0.001

Table 4 represents the number of genes selected by the methods and also shows the number of common genes identified between the methods. Diagonal elements are the number of genes selected by the methods and off diagonal elements are the common number of genes between themselves. To evaluate the performance of QRF we assumed 6^{-5} as the cut off point for the coefficient;s value of the genes. Number of genes selected by Quant-SIS-SCAD, Quant-SIS-lasso, Quant-SIS-MCP, Quant-tilting, Quant-PC, QRF and Hd-CQR are 9, 11, 11, 12, 14, 18 and 15 respectively. Here Quant-SIS-lasso and Quant-SIS-MCP select the same number of genes while the remaining methods select slightly higher number of genes. The Quant-PC identifies 14 genes while the common number of genes between Quant-PC and Quant-SIS-SCAD, Quant-SIS-lasso, Quant-SIS-MCP, Quant-Tilting is found as 7, 10, 10 and 8 respectively. The common genes that are identified by the methods include LC–24027, LC–33054, GATA binding protein 2, Thyroid Autoantigen 70kD, LC–31179.

Predictive performance of the methods are evaluated using the training models to predict the risk of death in test data sets. Mean square prediction error are calculated for each method using the fitted models displayed in Table 5. Patients are classified into high and low risk of death groups based on the median of fitted logarithm survival times. Fitted logarithm survival times larger than the median value are classified as lower risk group and lower ones hare classified as high risk group. The difference between logarithm survival times between two groups is tested with log-rank test.

Table 5 shows that the Quant-SIS-MCP outperforms all the methods in terms of MSE while Quant-Tilting and Quant-SIS-SCAD perform almost similar to the Quant-SIS-MCP. The QRF performs the worst than any other methods while the Quant-PC performs almost similar to the QRF. The Kaplan-Meier plots of the methods suggest that the groups are clearly separated based on high risk of death and lower risk of death.

4. Discussion and conclusion

n this research endeavor, we introduce innovative techniques for variable selection in the context of censored high-dimensional data through the utilization of quantile regression within Accelerated Failure Time (AFT) models. Our approach employs quantile regression principles to address AFT models, incorporating a suite of well established greedy algorithms from the literature–namely, sure independence screening (SIS), tilting, and PC-simple. These embedded greedy

algorithms play a pivotal role in accurately identifying significant and dynamic variables, ensuring a reliable variable selection process.

Our proposed methods are denoted as Quant-SIS, Quant-Tilting, and Quant-PC, with each method corresponding to a specific greedy algorithm (SIS, tilting, and PC-simple) integrated into the quantile regression framework for censored data. Additionally, we extend the Quant-SIS approach to three distinct variations–Quant-SIS-SCAD, Quant-SIS-Lasso, and Quant-SIS-MCP depending on the utilization of three optimization algorithms–SCAD, Lasso, and MCP to solve the SIS component.

To evaluate the performance of our proposed methods, we conduct comparative analyses against two established quantile-based methods found in existing literature: the quantile random forest (QRF) and high-dimensional censored quantile regression (CQR) methods. Through rigorous evaluation and comparison, we elucidate the effectiveness and applicability of our newly proposed techniques within the realm of censored high-dimensional data analysis.

Through comprehensive simulation studies and a real-world example involving Diffuse Large B-Cell Lymphoma (DLBCL) data, we unveil the potency of our proposed variable selection techniques within AFT models, particularly for high-dimensional censored datasets. The outcomes of our investigations reveal that the proposed techniques emerge as highly effective tools for variable selection, demonstrating comparable or even superior performance compared to the established quantile random forest method. As anticipated, the performances of our proposed methods exhibit minor fluctuations as the degree of censoring increases.

Furthermore, the assessment of prediction accuracy in terms of mean squared errors, based on the analysis of real DLBCL data, substantiates the applicability of our proposed methods as robust variable selection techniques for high-dimensional survival data scenarios. This underscores the potential utility of our quantile-based variable selection methods across a diverse array of fields, including but not limited to biostatistics, genomics, public health, and engineering. Hence, our proposed quantile based variable selection methods offer a versatile and effective means of detecting crucial variables or genes, serving as a valuable asset in the realm of high-dimensional data analysis.

We recognize that the screening process in the first stage of the two-stage method may influence variable selection in the second stage, potentially introducing bias and narrowing the generalizability of the method. To mitigate this, one could incorporate the cross-fitting to enhance the robustness of the two-stage method by efficiently utilizing the data while ensuring independence between the two stages.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

Agrawal, R., J. Gehrke, D. Gunopulos, and P. Raghavan. 1998. Automatic subspace clustering of high dimensional data for data mining applications. *ACM Sigmod Record* 27 (2):94–105.

Bang, H., and A. A. Tsiatis. 2002. Median regression with censored cost data. *Biometrics* 58 (3):643–9. doi: 10.1111/j.0006-341x.2002.00643.x.

Belloni, A., and V. Chernozhukov. 2011. l_1-penalized quantile regression in high-dimensional sparse models. *The Annals of Statistics* 39 (1):82–130. doi: 10.1214/10-AOS827.

Bühlmann, P., M. Kalisch, and M. H. Maathuis. 2010. Variable selection in high-dimensional linear models: Partially faithful distributions and the pc-simple algorithm. *Biometrika* 97 (2):261–78. doi: 10.1093/biomet/asq008.

Cho, H., and P. Fryzlewicz. 2012. High dimensional variable selection via tilting. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 74 (3):593–622. doi: 10.1111/j.1467-9868.2011.01023.x.

Efron, B. 1967. The two-sample problem with censored data. In *Porcessdings Fifth Berkeley Symposium in Mathematical Statistics* 4, eds. L. Le Cam and J. Neyman, 831–53. New York: Prentice Hall.

Fan, J. 1997. Comments on wavelets in statistics: A review by a. antoniadis. *Journal of the Italian Statistical Society* 6 (2):131–8. doi: 10.1007/BF03178906.

Fan, J., and R. Li. 2002. Variable selection for cox's proportional hazards model and frailty model. *The Annals of Statistics* 30 (1):74–99. doi: 10.1214/aos/1015362185.

Fan, J., and J. Lv. 2008. Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 70 (5):849–911. doi: 10.1111/j.1467-9868.2008.00674.x.

Faraggi, D., and R. Simon. 1998. Bayesian variable selection method for censored survival data. *Biometrics* 54 (4): 1475–85. doi: 10.2307/2533672.

Gui, J., and H. Li. 2005. Penalized cox regression analysis in the high-dimensional and low-sample size settings, with applications to microarray gene expression data. *Bioinformatics (Oxford, England)* 21 (13):3001–8. doi: 10.1093/bioinformatics/bti422.

He, X., L. Wang, and H. G. Hong. 2013. Quantile-adaptive model-free variable screening for high-dimensional heterogeneous data. *The Annals of Statistics* 41 (1):342–69. doi: 10.1214/13-AOS1087.

Johnson, B. A., D. Lin, and D. Zeng. 2008. Penalized estimating functions and variable selection in semiparametric regression models. *Journal of the American Statistical Association* 103 (482):672–80. doi: 10.1198/016214508000000184.

Khan, M. H. R. 2018. On the performance of adaptive pre-processing technique in analysing high-dimensional censored data. *Biometrical Journal* 60 (4):687–702. doi: 10.1002/bimj.201600256.

Khan, M. H. R., and M. Akhter. 2024. Ranking based variable selection for censored data using AFT models. *Communications in Statistics - Simulation and Computation* 53 (6):2917–39. doi: 10.1080/03610918.2022.2092639.

Khan, M. H. R., A. Bhadra, and H. Howlader. 2019. Stability selection for lasso, ridge and elastic net implemented with aft models. *Statistical Applications in Genetics and Molecular Biology* 18 (5):20170001. doi: 10.1515/sagmb-2017-0001.

Khan, M. H. R., and J. E. H. Shaw. 2016. Variable selection for survival data with a class of adaptive elastic net techniques. *Statistics and Computing* 26 (3):725–41. doi: 10.1007/s11222-015-9555-8.

Khan, M. H. R., and J. E. H. Shaw. 2019. Variable selection for accelerated lifetime models with synthesized estimation techniques. *Statistical Methods in Medical Research* 28 (3):937–52. doi: 10.1177/0962280217739522.

Khan, M. H. R. 2013. Variable selection and estimation procedures for high-dimensional survival data. Unpublished doctoral diss., University of Warwick.

Koenker, R., and G. Bassett. Jr, 1978. Regression quantiles. *Econometrica* 46 (1):33–50. doi: 10.2307/1913643.

Leng, C., and X. Tong. 2013. A quantile regression estimator for censored data. *Bernoulli* 19 (1):344–61. doi: 10.3150/11-BEJ388.

McLachlan, G. J., D. Peel, and R. Bean. 2003. Modelling high-dimensional data by mixtures of factor analyzers. *Computational Statistics & Data Analysis* 41 (3–4):379–88. doi: 10.1016/S0167-9473(02)00183-4.

Meinshausen, N. 2006. Quantile regression forests. *Journal of Machine Learning Research* 7 (Jun):983–99.

Park, S., X. He, and S. Zhou. 2017. Dantzig-type penalization for multiple quantile regression with high dimensional covariates. *Statistica Sinica* 27 (4):1619–38. doi: 10.5705/ss.202016.0081.

Peng, L., and Y. Huang. 2008. Survival analysis with quantile regression models. *Journal of the American Statistical Association* 103 (482):637–49. doi: 10.1198/016214508000000355.

Portnoy, S. 2003. Censored regression quantiles. *Journal of the American Statistical Association* 98 (464):1001–12. doi: 10.1198/016214503000000954.

Reich, B. J., and L. B. Smith. 2013. Bayesian quantile regression for censored data. *Biometrics* 69 (3):651–60. doi: 10.1111/biom.12053.

Rosenwald, A., G. Wright, W. C. Chan, J. M. Connors, E. Campo, R. I. Fisher, R. D. Gascoyne, H. K. Muller-Hermelink, E. B. Smeland, J. M. Giltnane, Lymphoma/Leukemia Molecular Profiling Project, et al. 2002. The use of molecular profiling to predict survival after chemotherapy for diffuse large-b-cell lymphoma. *The New England Journal of Medicine* 346 (25):1937–47. doi: 10.1056/NEJMoa012914.

Salerno, S., and Y. Li. 2023. High-dimensional survival analysis: Methods and applications. *Annual Review of Statistics and Its Application* 10 (1):25–49. doi: 10.1146/annurev-statistics-032921-022127.

Tibshirani, R. 1996. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 58 (1):267–88. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Tibshirani, R. 1997. The lasso method for variable selection in the cox model. *Statistics in Medicine* 16 (4):385–95. doi: 10.1002/(SICI)1097-0258(19970228)16:4<385::AID-SIM380>3.0.CO;2-3.

Tung, N. T., J. Z. Huang, T. T. Nguyen, and I. Khan. 2014. Bias-corrected quantile regression forests for high-dimensional data. In Proceedings of the 2014 International Conference on Machine Learning and Cybernetics, Lanzhou, 13, Vol. 1, pp. 1–6.

Wang, H., and L. Wang. 2009. Locally weighted censored quantile regression. *Journal of the American Statistical Association* 104 (487):1117–28. doi: 10.1198/jasa.2009.tm08230.
Wang, H. J., J. Zhou, and Y. Li. 2013. Variable selection for censored quantile regresion. *Statistical Sinica* 23 (1):145.
Wei, L.-J. 1992. The accelerated failure time model: A useful alternative to the cox regression model in survival analysis. *Statistics in Medicine* 11 (14-15):1871–9. doi: 10.1002/sim.4780111409.
Xia, X., B. Jian, J. Li, and W. Zhang. 2016. Low-dimensional confounder adjustment and high-dimensional penalized estimation for survival analysis. *Lifetime Data Analysis* 22 (4):547–69. doi: 10.1007/s10985-015-9350-z.
Xu, J., C. Leng, and Z. Ying. 2010. Rank-based variable selection with censored data. *Statistics and Computing* 20 (2):165–76. doi: 10.1007/s11222-009-9126-y.
Yu, H., and J. Yang. 2001. A direct lda algorithm for high-dimensional datawith application to face recognition. *Pattern Recognition* 34 (10):2067–70. doi: 10.1016/S0031-3203(00)00162-X.
Yu, L., and H. Liu. 2003. Feature selection for high-dimensional data: A fast correlation-based filter solution. In Proceedings of the 20th International Conference on Machine Learning (ICML-03), pp. 856–863.
Zhang, C. H. 2007. *Penalized linear unbiased selection*, 3. Department of Statistics and Bioinformatics, Rutgers University.
Zheng, Q., C. Gallagher, and K. Kulasekera. 2013. Adaptive penalized quantile regression for high dimensional data. *Journal of Statistical Planning and Inference* 143 (6):1029–38. doi: 10.1016/j.jspi.2012.12.009.
Zheng, Q., L. Peng, and X. He. 2015. Globally adaptive quantile regression with ultra-high dimensional data. *Annals of Statistics* 43 (5):2225–58. doi: 10.1214/15-AOS1340.
Zheng, Q., L. Peng, and X. He. 2018. High dimensional censored quantile regression. *Annals of Statistics* 46 (1): 308–43. doi: 10.1214/17-AOS1551.
Zou, H. 2006. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association* 101 (476):1418–29. doi: 10.1198/016214506000000735.
Zou, H., and T. Hastie. 2005. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 67 (2):301–20. doi: 10.1111/j.1467-9868.2005.00503.x.

Appendix

Appendix A. Additional Table: Example-I(b) and I(c)

Table A1. Summary results of example-1(b): $n = 200$ and $p = 200$, $p_\alpha = 5$ and $\tau = 0.25$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.000	0.000	0.000	0.000	0.020	0.000
	0	FP	0.000	0.001	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.020	0.001
		FN	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.001	0.000	0.000	0.000	0.002	0.005
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.002	0.005
50%		FN	0.000	0.000	0.000	0.010	0.000	0.000	0.001
	0	FP	0.000	0.002	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.002	0.000	0.010	0.000	0.000	0.002
		FN	0.000	0.010	0.000	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.010	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.010	0.020	0.000	0.000	0.000	0.001

Table A2. Summary results of Example-I(c): $n = 200$ and $p = 200$, $p_\alpha = 5$ and $\tau = 0.50$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.000	0.000	0.000	0.000	0.050	0.008
	0	FP	0.000	0.002	0.000	0.000	0.000	0.000	0.001
		FN+FP	0.000	0.002	0.000	0.000	0.000	0.050	0.009
		FN	0.000	0.000	0.000	0.000	0.000	0.000	0.008
	0.5	FP	0.000	0.001	0.000	0.000	0.000	0.007	0.001
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.007	0.009
50%		FN	0.000	0.001	0.000	0.020	0.000	0.000	0.009
	0	FP	0.000	0.002	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.003	0.000	0.020	0.000	0.000	0.009
		FN	0.020	0.006	0.010	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.000	0.000	0.000	0.000	0.008	0.008
		FN+FP	0.020	0.006	0.010	0.000	0.000	0.008	0.008

Appendix B. Additional Table: Example-II(b) and II(c)

Table B1. Summery results of example II(b): $n = 400$ and $p = 1000$, $p_\alpha = 10$ and $\tau = 0.25$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0	FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN	0.000	0.000	0.001	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.000	0.001	0.000	0.000	0.000	0.000
50%		FN	0.000	0.001	0.000	0.000	0.000	0.002	0.000
	0	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.002	0.000
		FN	0.000	0.001	0.000	0.000	0.000	0.001	0.000
	0.5	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.001	0.000

Appendix C. Additional Table: Example-I(d) with low SNR

Table C1. Summery results of example II(c): $n = 400$ and $p = 1000$, $p_\alpha = 10$ and $\tau = 0.50$.

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.000	0.000	0.000	0.000	0.000	0.000	0.001
	0	FP	0.000	0.001	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.000	0.001
		FN	0.000	0.000	0.001	0.000	0.000	0.000	0.000
	0.5	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.000	0.001	0.000	0.000	0.000	0.000
50%		FN	0.000	0.001	0.000	0.000	0.000	0.007	0.000
	0	FP	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.001	0.000	0.000	0.000	0.007	0.000
		FN	0.000	0.001	0.000	0.000	0.000	0.001	0.000
	0.5	FP	0.000	0.008	0.000	0.000	0.000	0.000	0.000
		FN+FP	0.000	0.009	0.000	0.000	0.000	0.001	0.000

Appendix D. Proof for properties of censored quantile regression

The objective function for censored quantile regression is given by

$$L(\beta(\tau), w) = \sum_{i=1}^{n} w_i \rho_\tau(Y_i - x_i^\top \beta(\tau)) + \sum_{i=1}^{n} (1 - w_i)\rho_\tau(Y^{+\infty} - x_i^\top \beta(\tau)),$$

where

- $\rho_\tau(u) = u(\tau - \mathbb{I}(u < 0))$ is the quantile check function.
- w_i are weights defined by:

$$w_i = \begin{cases} 1, & \text{if } \delta_i = 1 \text{ (uncensored)}, \\ 0, & \text{if } \delta_i = 0 \text{ and } \pi_i > \tau, \\ \dfrac{\tau - \pi_i}{1 - \pi_i}, & \text{if } \delta_i = 0 \text{ and } \pi_i \le \tau, \end{cases}$$

where $\pi_i = P(T_i \le C_i | x_i)$ is the conditional survival probability.

- $Y^{+\infty}$ is a sufficiently large value representing censored contributions.

The following conditions are required for the properties of censored quantile regression:

1. The censoring mechanism is independent of the event time given covariates:

$$(T_i \perp C_i | x_i).$$

   This ensures that the censoring process does not bias the quantile estimation.
2. The conditional survival function $F(C_i|x_i)$ is smooth and well-behaved. This is necessary for accurate redistribution of the censored observations.
3. The covariate matrix X has full rank, ensuring the identifiability of the quantile regression coefficients $\beta(\tau)$.

Proof of Consistency of $\hat{\beta}(\tau)$

Under the independent censoring assumption, the expected value of the objective function converges to the population loss:

$$\mathbb{E}[L(\beta(\tau), w)] = \mathbb{E}\big[W\rho_\tau(Y - x^\top \beta(\tau)) + (1 - W)G_\tau(Y|X)\big],$$

where:

- $W = w_i$ is the weight for observation i, incorporating censoring information.
- $G_\tau(Y|X)$ represents the contribution of censored observations, derived from the redistribution of mass.

Table D1. Summary results of Example-I: $n = 200$ and $p = 200$, $p_\alpha = 5$ and $\tau = 0.60$ with low SNR ($\beta = 5$ and $\sigma = 2$).

$P_{\%}$	r	Rates	Quant − SIS − SCAD	Quant − SIS − Lasso	Quant − SIS − MCP	Quan $t-$ tilting	Quant − PC	QRF	Hd-CQR
30%		FN	0.001	0.005	0.003	0.002	0.002	0.100	0.018
	0	FP	0.001	0.007	0.002	0.002	0.003	0.006	0.007
		FN+FP	0.002	0.012	0.005	0.004	0.005	0.106	0.025
		FN	0.002	0.007	0.005	0.003	0.004	0.110	0.022
	0.5	FP	0.003	0.015	0.006	0.004	0.003	0.012	0.008
		FN+FP	0.010	0.022	0.011	0.007	0.007	0.122	0.030
50%		FN	0.001	0.007	0.06	0.006	0.004	0.112	0.021
	0	FP	0.002	0.007	0.003	0.005	0.004	0.011	0.010
		FN+FP	0.003	0.014	0.009	0.011	0.008	0.123	0.031
		FN	0.060	0.030	0.035	0.004	0.005	0.007	0.008
	0.5	FP	0.015	0.012	0.005	0.004	0.006	0.030	0.025
		FN+FP	0.075	0.042	0.040	0.008	0.011	0.038	0.033

The estimator $\hat{\beta}(\tau)$, which minimizes the empirical loss $L(\beta(\tau), w)$, converges in probability to the true quantile coefficient:

$$\hat{\beta}(\tau) \xrightarrow{p} \beta(\tau), \quad \text{as } n \to \infty.$$

This result follows from the uniform convergence of the empirical distribution to the population distribution, a direct consequence of the independent censoring assumption and regularity conditions.

Proof of Asymptotic Normality
The first-order conditions for the weighted loss function are:

$$\frac{\partial L(\beta(\tau), w)}{\partial \beta} = \sum_{i=1}^{n} \left[w_i \psi_\tau(Y_i - x_i^\top \beta(\tau)) + (1 - w_i) \psi_\tau(Y^{+\infty} - x_i^\top \beta(\tau)) \right] = 0,$$

where $\psi_\tau(u) = \tau - \mathbb{I}(u < 0)$ is the derivative of the check function.

Under the regularity conditions–the design matrix X has full rank, the weights w_i are consistent estimators of the true censoring probabilities, the check function $\rho_\tau(u)$ is Lipschitz continuous, the estimator $\hat{\beta}(\tau)$ satisfies the asymptotic normality property:

$$\sqrt{n}(\hat{\beta}(\tau) - \beta(\tau)) \xrightarrow{d} N(0, \Sigma(\tau)),$$

where the asymptotic covariance matrix $\Sigma(\tau)$ is given by:

$$\Sigma(\tau) = \tau(1 - \tau)(X^\top W X)^{-1},$$

and W is the weight matrix incorporating both uncensored and censored contributions.

Appendix E. Derivative of the SCAD Penalty Function

The SCAD penalty function is defined as:

$$p_\lambda(|\beta(\tau)|) = \lambda \left\{ I(|\beta(\tau)| \le \lambda) + \frac{(a\lambda - |\beta(\tau)|)_+}{(a-1)\lambda} I(|\beta(\tau)| > \lambda) \right\}, \quad \text{for some } a > 2.$$

Derivative of the Penalty Term:

1. When $|\beta(\tau)| \le \lambda$:
   In this region, the penalty function is linear, and the derivative is:
   $$\frac{\partial p_\lambda(|\beta(\tau)|)}{\partial \beta(\tau)} = 0.$$
2. When $|\beta(\tau)| > \lambda$:
   The penalty function transitions to:
   $$\frac{(a\lambda - |\beta(\tau)|)}{(a-1)\lambda}.$$
   The derivative in this case is:
   $$\frac{\partial p_\lambda(|\beta(\tau)|)}{\partial \beta(\tau)} = -\frac{\text{sgn}(\beta(\tau))}{(a-1)\lambda},$$

where $\text{sgn}(\beta(\tau))$ is the sign function, defined as:

$$\text{sgn}(\beta(\tau)) = \begin{cases} 1, & \beta(\tau) > 0, \\ -1, & \beta(\tau) < 0. \end{cases}$$

3. Combined Result:
The derivative of the SCAD penalty term is:

$$\frac{\partial p_\lambda(|\beta(\tau)|)}{\partial \beta(\tau)} = \begin{cases} 0, & |\beta(\tau)| \leq \lambda, \\ -\dfrac{\text{sgn}(\beta(\tau))}{(a-1)\lambda}, & |\beta(\tau)| > \lambda. \end{cases}.$$